

Does spatial resolution explain the MERRA-2 vs. MODSCAG disagreement?

EXPLORATORY FOLLOW-UP · NO NEW DATA PULLED

Question: how much of the pooled bias reflects MERRA's coarse $0.625^\circ \times 0.5^\circ$ grid vs. some other cause?

Constraint: reuse existing outputs only — no reprocessing/redownload, no change to the aggregation rule (MODIS up to MERRA; never MERRA down).

Terrain heterogeneity proxy: bin the checked-in USGS 3DEP DEM (800×600 px) into the 72 MERRA cells; within-cell elevation std. dev. — steep, heterogeneous cells are worst represented by a coarse grid.

Bias signal: recombine already-computed monthly sufficient stats (pixel_stats.csv climatology-month rows) per cell into pooled WY2010–2023 bias/NMB, same 5% low-snow mask as the public figure.

Two comparisons: (1) terrain heterogeneity vs. spatial bias; (2) spatial vs. temporal bias variance.

Inputs (all pre-existing)

[tests/fixtures/domain_dem_3dep.tif](#)

real 3DEP DEM, checked in

[results/water_year_2010_2023_pixel_stats.csv](#)

per-cell climatology sufficient stats

[results/water_year_2010_2023_overall_stats.csv](#)

per-year domain sufficient stats

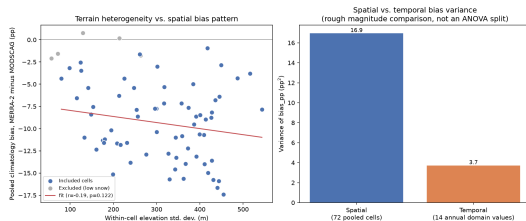
New, exploratory-only artifact:

[scripts/resolution_attribution.py](#) →

[results/derived_resolution_attribution.{csv,png}](#)

Result: bias pattern is mostly not explained by terrain heterogeneity

FINDINGS



Correlation: $r = -0.19$ ($p = 0.12$, $R^2 \approx 0.04$) across 67 of 72 cells (5 masked for low snow) — essentially no relationship, and the weak trend runs opposite to a resolution-mismatch story.

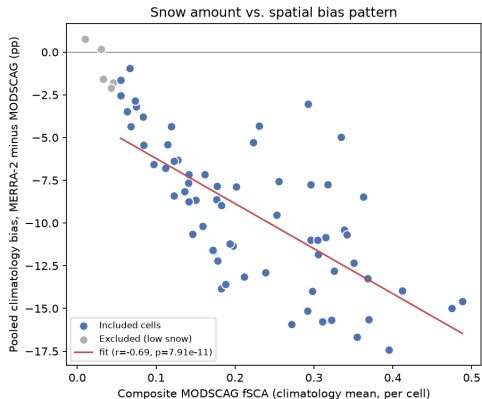
Variance split: spatial variance across cells ($\approx 16.9 \text{ pp}^2$) is $\sim 4.5\times$ the interannual variance at the domain level ($\approx 3.7 \text{ pp}^2$) — the disagreement is mostly a stable spatial pattern, not year-to-year.

Interpretation: spatial pattern doesn't track terrain heterogeneity, so resolution/aggregation mismatch looks like a minor contributor. Next: land cover, radiation/snow-physics differences, or a location-dependent MERRA bias unrelated to subgrid relief.

Correlational, not a controlled regridding experiment — MERRA is never resampled to MODIS resolution in this repository. See [results/derived_resolution_attribution.csv](#) for full per-cell values.

Bias tracks how much snow a cell has — but that doesn't isolate resolution

FOLLOW-UP · DOES SNOW AMOUNT EXPLAIN IT INSTEAD?



Motivating idea: finer MODSCAG pixels can resolve a coarse MERRA cell that is partly snow, partly bare; MERRA represents it with one value. Disagreement should be worst in *mixed-state* cells, proxied by $fSCA \times (1 - fSCA)$, peaking at $fSCA = 0.5$.

What the data show: every included cell has composite $fSCA < 0.5$ (0.05–0.49) — on this side of 0.5, $fSCA \times (1 - fSCA)$ is monotonic in $fSCA$, so it can't be told apart from raw snow amount here. Raw $fSCA$ vs. bias: $r = -0.69$ ($R^2 \approx 0.48$); mixed-state proxy: $r = -0.72$ ($R^2 \approx 0.52$) — nearly the same.

Honest read: more snow $\Rightarrow$ more negative bias is real and strong, but can't separate a resolution/mixed-pixel mechanism from other snow-amount-scaling effects (e.g. a MERRA depletion-curve or physics bias unrelated to pixel size).

A direct mixed-pixel test needs within-cell fine-pixel variance, not just the mean — not recoverable from saved sufficient statistics without reprocessing daily granules.